

Heidi Pamela Martínez Martínez
1952947 N1

Actividad 3.5

Máquina	Operador						Σ	$k=6$ $n=24$
	1	2	3	4	5	6		
1	42.5	39.3	39.6	39.9	42.9	43.6	247.8	$C = \frac{1010.9}{24}$
2	39.8	40.1	40.5	42.3	42.5	43.1	248.3	
3	40.2	40.5	41.3	43.4	44.9	45.1	255.4	$C = 42\ 579.95$
4	41.3	42.2	43.5	44.2	45.9	42.3	259.4	
Σ	163.8	162.1	164.9	169.8	176.2	174.1	1010.9	

$$SCE = 81.86 - 15.93 - 42.09 = \underline{23.84} \quad f_T = \frac{5.31}{1.59} = \underline{3.34} \quad f_B = \frac{8.42}{1.59} = \underline{5.29}$$

$$STC = (42.5^2 + 39.8^2 + 40.2^2 + 41.3^2 + 39.3^2 + 40.1^2 + 40.5^2 + 42.2^2 + 39.6^2 + 40.5^2 + 41.3^2 + 43.5^2 + 39.9^2 + 42.3^2 + 43.4^2 + 44.2^2 + 42.9^2 + 42.5^2 + 44.9^2 + 45.9^2 + 43.6^2 + 43.1^2 + 45.1^2 + 42.3^2) - 42\ 579.95 = \underline{81.86}$$

$$SCB = \left(\frac{163.8^2}{4} + \frac{162.1^2}{4} + \frac{164.9^2}{4} + \frac{169.8^2}{4} + \frac{176.2^2}{4} + \frac{174.1^2}{4} \right) - 42\ 579.95 = \underline{42.09}$$

$$SCT = \left(\frac{247.8^2}{6} + \frac{248.3^2}{6} + \frac{255.4^2}{6} + \frac{259.4^2}{6} \right) - 42\ 579.95 = \underline{15.93}$$

$$CMB = \frac{42.09}{5} = \underline{8.42} \quad CMT = \frac{15.93}{3} = \underline{5.31} \quad CME = \frac{23.84}{15} = \underline{1.59}$$

Var	Suma de cuadrados	Grados de libertad	Cuadrado medio	F
Máquina	15.93	3	5.31	3.34
Operaciones	42.09	5	8.42	5.29
Errores	23.84	15	1.59	
Total	81.86	23		

Respuesta ↓

$$f_c = 3.34 > f_T = 3.29$$

Si se rechaza H_0 ya que no hay igualdad entre las medias